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Housing for drain outlet

Abstract

An apparatus is provided for protecting from freezing fluid flowing from a conduit outlet of a condensing gas furnace above ground level on an exterior wall of a building. The freeze protection apparatus comprises a heater box disposed within a housing, including thermal insulating material lining the interior surface of walls of the housing. A thermostatically controlled heater is mounted to the heater box for maintaining the interior chamber above freezing temperature to prevent condensate fluid from freezing. Means are provided for mounting the housing to structure adjacent the conduit outlet. Heat is retained within the housing to prevent condensate fluid within the conduit outlet from freezing.

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Background/Summary

BACKGROUND

(1) This invention relates generally to a housing for a termination point of a drain line outlet associated with all central heating and air conditioning systems. The housing is also configured to accommodate the traps on all condensing furnaces, as well as a condensate pump that mechanically

expels liquid condensate when required. More particularly, the housing or enclosure is heated for use with a condensing gas furnace is those areas where the condensate is susceptible to freezing. (2) Condensing gas furnaces are gas-fired furnaces having annual fuel efficiency ratings of 90% or above. Due to the nature of their twin heat exchanger design and ultra-high efficiency ratings, condensing furnaces can produce three to seven gallons of condensate fluid daily during a heat cycle. Moreover, the condensing furnace traps require condensate pumps when the furnace is situated below grade elevations. Condensate drain outlets are all necessary components for expelling condensate liquid produced by the furnace to an exterior location beyond the home or structure's perimeter. However, during extremely cold weather conditions, the condensate fluid is prone to freezing at the traps, the condensate pump, or the drain outlets. Windy conditions significantly contribute to increased chill factors that cause condensate freezing at the exterior drain outlet termination points. The condensate traps condensate pumps both have differed size reservoirs that hold small quantities of standing condensate fluid which is especially susceptible to freezing in frigid conditions in and around the furnace locations, such as below grade basements or crawlspaces.

(3) Once condensate fluid freezes at any location, sensitive fail-safe devices on the furnaces immediately shut the furnace off, resulting in a "no-heat" scenario. This situation typically requires a service call and a technician to come out, verify which component is frozen, and then subsequently thaw the frozen components. Unfortunately, this is a temporary fix as the drain line components may freeze again depending upon the ambient conditions, leaving the home's occupants with no-heat when they need it the most.

(4) Applicant is the owner of U.S. Pat. No. 9,469,976, entitled Housing For Drain Outlet, which describes a solution to prevent freezing of the condensate at the drain outlet termination point. The contents of U.S. Pat. No. 9,469,976, are incorporated herein by reference in the entirety. The '976 patent describes an apparatus for protecting condensate fluid from a condensing gas furnace from freezing at an above-ground conduit outlet on an exterior wall of a home or building. The freeze protection apparatus comprises a housing adapted for covering the conduit outlet. The housing includes a top wall, a front wall, and a pair of opposed side walls interconnecting the top wall and the front wall. The housing partially defines an enclosed area having a rear aperture and a bottom aperture. Thermal foam insulating panel material lines the interior surface of the walls of the housing. Means are provided for mounting the housing to the exterior wall, fully encompassing the conduit outlet. The inner edges of the side walls and the top wall contact the exterior wall, and the bottom edges of the side walls and the front wall contact the ground below the conduit outlet for enclosing the conduit outlet within an enclosed space defined by the housing, exterior wall and the ground. Heat is retained within the housing to prevent fluid within the conduit from freezing. The patented freeze protection apparatus has proven to be a valuable asset as the use of condensing gas furnaces continues to grow and will be mandated by the Department of Energy in January 2029. Thus, the problem of condensate fluid freezing will become more prevalent, especially in consistently colder environments.

(5) For the foregoing reasons, a freeze protection apparatus is needed for condensing gas furnace drainage components. Ideally, the freeze protection apparatus will include a housing to prevent freezing of the condensate fluid exiting the furnace at any location. The new housing should be equipped with an auxiliary heat component to prevent freezing in particularly cold environments. Ideally, the housing will allow for relatively quick and easy installation or replacement.

SUMMARY

(6) An apparatus is provided for protecting from freezing fluid flowing from a conduit outlet above ground level on an exterior wall of a building. The freeze protection apparatus comprises a condensing gas furnace providing a continuous flow of condensate fluid through the conduit outlet. A heater box has a longitudinal dimension and includes a top wall, a front wall and a plurality of side walls interconnecting the top wall and the front wall. The heater box partially defines an

enclosed area sized for covering the conduit outlet and having a rear aperture and a bottom aperture. A housing is adapted for receiving the heater box. The housing includes a top wall having an interior surface and an inner edge, a front wall having an interior surface, a bottom edge, and a lateral dimension, and a pair of opposed side walls each side wall having an interior surface, an inner edge, a bottom edge and a depth dimension. The side walls interconnect the top wall and the front wall. The housing has a longitudinal dimension greater than the longitudinal dimension of the heater box and partially defines an enclosed area having openings at a rear aperture for passing the heater box. Thermal insulating material lines the interior surface of the walls of the housing. A thermostatically controlled heater mounted to the heater box maintains the interior chamber above freezing temperature to prevent fluid from freezing. Means are provided for mounting the housing to the exterior wall adjacent the conduit outlet. Heat is retained within the heater box to prevent fluid within the conduit from freezing.

(7) In one aspect, the freeze protection apparatus may further comprise a bottom wall spanning the side walls and the front wall inwardly longitudinally spaced from distal edges of the side walls and the front wall. The bottom wall may be perforated to allow fluid to pass through.

(8) In another aspect, the electric heater comprises a silicone pad heater.

(9) In use, the inner edges of the side walls and the top wall of the housing contact the exterior wall. The bottom edges of the side walls and the front wall may also contact the ground at ground level below the conduit outlet for enclosing the conduit outlet within the enclosed area defined by the heater box, the exterior wall and the ground level.

(10) In one embodiment, the mounting means comprises flanges at the inner edges of the side wall and the top wall of the housing. In another embodiment, the mounting means comprises a hinge for hingedly securing the housing to the exterior wall for pivoting movement between an open position away from the conduit outlet and a closed position enclosing the conduit outlet. In one aspect, the pivot axis of the housing is about a substantially horizontal axis.

(11) An apparatus is also for protecting from freezing fluid flowing from a condensing gas furnace providing a continuous flow of condensate fluid through a trap. The freeze protection apparatus comprises a heater box having a longitudinal dimension. The heater box includes a top wall, a front wall, and a plurality of side walls interconnecting the top wall and the front wall. The heater box partially defines an enclosed area and has a rear aperture and a bottom aperture for receiving the trap into the enclosed area. A housing receives the heater box, the housing including a top wall having an interior surface and an inner edge, a front wall having an interior surface, a bottom edge and a lateral dimension, and a plurality of side walls, each side wall having an interior surface, an inner edge, a bottom edge and a depth dimension. The side walls interconnect the top wall and the front wall such that the housing partially defines an enclosed area having openings at a rear aperture and a bottom aperture. The area of the bottom aperture defined by the lateral dimension of the front wall and the depth dimension of the side walls is configured for receiving the heater box. A thermal insulating material lines the interior surface of the walls of the housing. A thermostatically controlled electric heater is mounted to the heater box for maintaining the temperature of the enclosed area above freezing temperature to prevent the condensate fluid from freezing. Means are provided for mounting the housing to the furnace adjacent the trap. The inner edges of the side walls and the top wall of the housing contact the furnace for enclosing the trap within an enclosed space defined by the heater box. Heat is retained within the heater box to prevent the condensate fluid from freezing.

(12) In one aspect, the freeze protection apparatus may further comprise a bottom wall spanning the side walls and the front wall inwardly longitudinally spaced from distal edges of the side walls and the front wall. The bottom wall may be perforated to allow fluid to pass through.

(13) In another aspect, the electric heater comprises a silicone pad heater.

(14) In one embodiment, the mounting means comprises flanges at the inner edges of the side wall and the top wall of the housing. In another embodiment, the mounting means comprises a hinge for

hingedly securing the housing to the exterior wall for pivoting movement between an open position away from the conduit outlet and a closed position enclosing the conduit outlet. In one aspect, the pivot axis of the housing is about a substantially horizontal axis.

(15) Yet another apparatus is provided for protecting from freezing fluid flowing from a condensate pump in a condensing gas furnace providing a continuous flow of condensate fluid. The freeze protection apparatus comprises a first pan having a depth and a longitudinal dimension, the first pan configured for receiving the condensate pump. A second pan has a depth and a longitudinal dimension greater than the longitudinal dimension of the first pan for receiving the first pan. A thermostatically controlled electric heater is mounted to the second pan for maintaining condensate fluid in the first pan and the second pan above freezing temperature to prevent the fluid from freezing. In one embodiment, the electric heater comprises a silicone pad heater.

(16) In one aspect, the freeze protection apparatus further comprises outwardly depending flanges at the upper edges of the first pan for suspending the first pan in the second pan.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present invention, reference should now be had to the embodiments shown in the accompanying drawings and described below. In the drawings:

(2) FIG. 1 is a front perspective view of an embodiment of a housing with auxiliary heat for accommodating a drain outlet.

(3) FIG. 2 is a front elevation view of the housing as shown in FIG. 1.

(4) FIG. 3 is a bottom rear perspective view of the housing as shown in FIG. 1.

(5) FIG. 4 is a top plan view of the housing as shown in FIG. 1.

(6) FIG. 5 is a bottom plan view of the housing as shown in FIG. 1.

(7) FIG. 6 is a rear elevation view of the housing as shown in FIG. 1.

(8) FIG. 7 is a rear elevation view of the housing with a cover removed exposing an inside of an electronics compartment.

(9) FIG. 8 is an exploded rear perspective view of the housing as shown in FIG. 1.

(10) FIG. 9 is a right side elevation view of the housing as shown in FIG. 1.

(11) FIG. 10 is a left side elevation view of the housing as shown in FIG. 1.

(12) FIG. 11 is a side elevation view of the housing as shown in FIG. 1 installed on an exterior wall over a drain outlet in a first open position and a second closed position shown in phantom.

(13) FIG. 12 is a top perspective view of the housing as shown in FIG. 1 installed on the exterior wall over the drain outlet as shown in FIG. 10.

(14) FIG. 13 is a schematic view of electrical wiring for powering auxiliary heat in the housing as shown in FIG. 1.

(15) FIG. 14 is a schematic view of another embodiment of a housing for accommodating a drain trap installed at a trap location for a horizontal condensing gas furnace.

(16) FIG. 15 is a schematic view of the housing as shown in FIG. 14 installed at a trap location for a vertical condensing gas furnace.

(17) FIG. 16 is a schematic view of a third embodiment of a housing for accommodating a condensate fluid pump installed as a heated compartment for a condensate fluid pump of a condensing gas furnace.

(18) FIG. 17 is a schematic side elevation view of an assembly including the housing as shown in FIG. 16 with the compartment for receiving a portion of the condensate fluid pump.

(19) FIG. 18 is an exploded side perspective view of the housing assembly as shown in FIG. 17.

(20) FIG. 19 is a partially exploded side perspective view of another embodiment of a housing for receiving a condensate fluid pump as shown in FIG. 16 including a drop-in metal sleeve for

supporting the pump assembly.

DESCRIPTION

(21) Certain terminology is used herein for convenience only and is not to be taken as a limitation on the invention. For example, words such as “upper,” “lower,” “left,” “right,” “horizontal,” “vertical,” “upward,” “transverse” and “downward” merely describe the configuration shown in the FIGS. Indeed, the components may be oriented in any direction and the terminology, therefore, should be understood as encompassing such variations unless specified otherwise.

(22) Referring now to the drawings, wherein like reference numerals designate corresponding or similar elements throughout the several views, an embodiment of a housing assembly for accommodating a drain outlet is shown in FIGS. **1-10** and generally designated at **20**. The housing assembly **20** comprises a housing **22** for accommodating a drain outlet and means for removably securing the housing **22** to a supporting surface **25**, such as an exterior wall or a foundation of a house or to a furnace itself.

(23) The housing **22** comprises a plurality of walls, including a right side wall **26** and a left side wall **28**, a front wall **30**, and a top wall **32** extending transversely between and interconnecting the side walls **26**, **28** and the front wall **30**. The plurality of walls **26**, **28**, **30**, **32** extend generally continuously from their respective adjoining edges. In the configuration shown, the side walls **26**, **28** and the front wall **30** taper outwardly from the top wall **32** to their distal edges. A perforated bottom wall **33** spans the side walls **26**, **28** and the front wall **30** inwardly spaced from their distal edges. The housing **22** shown in FIGS. **1-10** defines a rear opening **34**.

(24) The walls **26**, **28**, **30**, **32**, **33** of the housing may be generally planar and may take on any shape, such as a substantially rectangular or rhomboid shape as shown. Preferably, the walls **26**, **28**, **30**, **32**, **33** are shaped so as to conform to the shape of the supporting surface **25** from which the housing **22** may be suspended. Each wall **26**, **28**, **30**, **32**, **33** includes an outwardly facing outer surface and an inwardly facing inner surface. The inner surfaces of the walls **26**, **28**, **30**, **32**, **33** at least partially define an interior chamber of the housing assembly **20**, which is further defined by the support surface **25** to which the housing **22** is mounted. The walls **26**, **28**, **30**, **32**, **33** of the housing **22** may be fabricated from any material, including, but not limited to, any sheet metal, (e.g. steel, aluminum, galvanized metal, etc.), plastic, fiberglass, metal or any other type of rigid material. The material of the housing **22** should be able to withstand weather, as well as being hit, bumped or other damage typical around the house. While the housing **22** can be assembled by fastening the separate walls **26**, **28**, **30**, **32** to one another (e.g., as by welding, L-bracket, fasteners, etc.), the housing **22** may be a single integral sheet folded along fold lines. It is understood that the housing **22** may be of any appropriate dimension for accommodating the drain outlet.

(25) A heater box **60** is configured to be received within the interior chamber of the housing **22**. In one embodiment, the heater box **60** is similarly shaped to the housing **22**. As such, the heater box **60** comprises a plurality of walls, including a right side wall **62** and a left side wall **64**, a front wall **66**, and a top wall **68** extending transversely between and interconnecting the side walls **62**, **64** and the front wall **66** along their edges. In the embodiment best shown in FIG. **8**, the configuration of the heater box **60** is shorter than housing **22** while generally corresponding to the shape of the housing **22**. Accordingly, the side walls **62**, **64** and the front wall **66** of the heater box **60** taper outwardly from the top wall **68** to their distal edges and define an interior **70** opening to the rear. When the heater box **60** is received within the housing **22**, an electrical compartment is formed between the top wall **32** of the housing **22** and the top wall **68** of the heater box **60**.

(26) A single layer of foam insulation board **37** lines the inner surface of each of the walls of the housing **22**, except for the perforated bottom wall **33**. When assembled, the foam insulation panels **37** are sandwiched between the inner surface of each of the walls of the housing **22** and the outer surface of each of the walls of the heater box **60**, except for the perforated bottom wall **33**. The foam insulation panels **37** can be attached to the interior surface of the walls **26**, **28**, **30**, **32** of the housing **22** using a standard, commercial-grade “brush-on” or “spray-on” construction adhesive.

Suitable foam insulation boards or panels are about $\frac{3}{8}$ inches thick and have an R-factor of at least about three (R-3). Other types of insulation may be used, including rotary liner insulation, attached with similar brush-on or spray-on aerosol adhesive or individual “weld-pins.” The type and thickness of the insulation, whether foam panel board, spray-on foam, or rotary liner, may be selected as is suitable for the geographical area in which the housing assembly **20** is being used.

(27) The mounting means **24** comprises side flanges **38**, extending perpendicularly outwardly from the rear edge of each of the side walls **26**, **28** and a top flange **39** extending perpendicularly outwardly from the rear edge of the top wall **32**. Each of the side flanges **38** and the top flange **39** may have one or more openings **41** for receiving a fastener **40**, e.g., bolts, rods, screws, hooks, nails, etc., for attaching the housing **22** to the support surface **25**. The openings **41** may be of any shape, such as a round hole, a keyhole slot, eyelet-shape, etc.

(28) The top flange **39** may comprise a hinge **42** for connecting the housing **22** via the top wall **32** to the support surface **25**. One side of the hinge **42** is pivotally connected to a portion of the top wall **32** of the housing **22**. The hinge **42** may be removeably fastened (e.g., by fasteners, screws, bolts, etc.) or may be permanently fixed (e.g., as by welding) to the top wall. The other side of the hinge **42** is connected to the support surface **25** (FIGS. **11** and **12**), which allows the housing **22** to be rotated away from the support structure when necessary. Any of the side walls **26**, **28** of the housing **22** may also comprise a hinge. It is understood that the mounting means **24**, as shown and described herein, is not limited to the embodiments. Any other suitable means may be used for attaching the housing **22** to the supporting structure **25**, including magnets and the like.

(29) Referring to FIGS. **6-8**, an electric heater **50** is provided for delivering auxiliary heat when required by environmental conditions to prevent freezing of the condensate fluid from the condensing gas furnace. In one embodiment, the heater **50** comprises a silicone pad heater secured to the inner surface of the front wall **30** by a perforated metal cover **52** fastened to the front wall **30**. The pad heater **50** is sandwiched between the front wall **30** and the cover **52**. As best seen in FIG. **7**, a compartment **54** for electronic components is formed between the top wall **32** of the housing **22** and the top wall **68** of the heater box **60**. The latter provides a planar metal floor spanning the front wall **30** and side walls **26**, **28** of the housing **22** spaced from the top wall **32**. A standard three-wire prong plug and cord **58** provides electrical power to the housing **22**. The cord **58** passes through the right side wall **26** of the housing **22** and into the compartment **54** for delivering the power to the electronic components. A planar compartment cover **66** is provided for enclosing the electronics components within the compartment **54**. The cover **66** may be selectively removable (FIGS. **7** and **8**) by adhering Velcro **67** to the flanges **38**, **39** and the inner surface of the cover **66**.

(30) As shown in FIG. **13**, the electronic components comprise a thermostat **80**, an on/off indicator light **82** on the outer surface of the top wall **32** of the housing **22**, and a ground plug **84**. A hot wire **90** and a neutral wire **92** are each electrically connected to the thermostat **80** and the light **82**. A ground wire **85** from the cord **58** is secured to the ground plug **84**. The thermostat **80** functions to actuate the heater **50** for maintaining the interior of the housing **22** above the freezing temperature of the condensate fluid. In one embodiment, the heater **50** maintains the interior of the housing **22** at a temperature of between about 42 degrees and 80 degrees.

(31) In use, the housing assembly **20** is secured to a supporting structure **25** surrounding a drain outlet **44** (FIGS. **11** and **12**). The housing **22** is positioned with the rear opening of the housing **22** adjacent to the support surface **25** such that each flange **38**, **39** of the mounting means **24** engages the support surface **25**. Fasteners **40** are inserted through the holes **41** in the flanges **38**, **39** and into the support surface **25**, as shown. This effectively mounts the housing **22** to the support surface **25**, which abuts against the rear edges of the side walls **26**, **28**, the top wall **30** and the flanges **38**. The housing **22** is shown resting on the ground for further supporting the weight of the housing. However, it is understood that the housing may be suspended above the ground, depending upon where the drain outlet exits the support surface **25** structure. The interior chamber defined between the surface of the supporting **25** surface and the inner surfaces of the walls **62**, **64**, **66**, **68** of the

heater box **60** accommodates the drain outlet **44**. The extension of the side walls **62**, **64** from the support surface **42** determines the depth of the interior chamber defined by the heater box **60**. The internal space is at least large enough for receiving the drain outlet. As best seen in FIG. **11** in phantom, the internal space is not accessible via the rear opening **24** or the bottom opening **36** of the housing **22**, the former being covered by the support surface **25** and the latter being closed to the ground (FIG. **11**) when the housing **22** contacts the ground. When the drain outlet **44** is covered, the fluid will run from the drain outlet **44** to the ground. The thermostatically controlled heater **50** will maintain the interior chamber above freezing temperatures to prevent risk of freezing of the condensate fluid under frigid conditions. Mounting the housing **22** around the drain outlet **44** provides an insulated, heated internal space to protect the drain outlet **44** and prevent freezing of the draining fluid.

(32) At any time, such as for maintenance, replacement, upgrades, etc., the housing **22** may be detached from the support surface **25** via the mounting means **24**, as previously described with regard to FIGS. **11** and **12**. Allowing the housing assembly **20** to be easily and quickly removed from the support surface **25** gives free access to the housing **22** of the drain outlet **44** for performing various operations after installation, such as performing repair work, upgrading systems or maintenance or replacing the housing altogether.

(33) FIG. **11** shows one drain outlet **44**; however, it should be understood that the housing **22** may accommodate more or less outlets than what is shown. In the embodiments shown, the drain outlet **44** is representative of a drain outlet for a condensing furnace. As shown in FIG. **12**, the housing assembly **20** allows mulch to be placed over the top of the drain line, further restricting wind or extreme temps from accessing the drain outlet. The housing assembly **20** also prevents insects, such as wasps, from colonizing the end of the drain outlet, thus blocking the flow of condensate, and again cutting the HVAC system off. The housing assembly **20** also prevents rodents such as squirrels from gnawing away the exposed portion of the drain outline, which would allow the condensate to run into the exterior wall cavities rather than to the exterior. Protecting the (typically plastic and somewhat fragile) exposed drain outlet from damage from mowers, weed eating, and prevention of mulching and annual re-mulching from clogging up end of condensate drain outlet is also a benefit of the drain housing.

(34) The housing assembly **20** may be used in cooperation with other locations of a condensing gas furnace. In one embodiment, shown in FIGS. **14** and **15**, the housing assembly **20** may protect fluid from freezing in a trap location **94** of a condensing gas furnace. The trap **94** in a condensing gas furnace serves as the primary evacuation device for removing condensate from the furnace. A trap **94** location for a horizontal condensing gas furnace **90** and a trap location for a vertical condensing gas furnace **92** are shown in FIGS. **14** and **15**, respectively. The housing assembly **20** may be used to surround the trap **94** below the horizontal furnace **90** or the trap **94** on the side of the vertical furnace **92** to prevent condensate fluid from freezing.

(35) A condensing gas furnace **90** may also have a condensate pump assembly **70** (FIGS. **16** and **17**) to pump liquid from where the furnace is located in a building to outside of the building. An embodiment of the present housing assembly **20** can be used to prevent fluid in the condensate pump assembly **70** from freezing. FIG. **16** shows the relative position of a condensate pump assembly **70** below the horizontal condensing gas furnace **90**. A close-up view of the condensate pump assembly **70** received within the heating assembly **20** is shown in FIG. **17**. The housing assembly **20** for this application is shown in FIG. **18**. The housing assembly **20** generally comprises all of the same components as the first embodiment (FIGS. **1-10**), but is configured in a generally rectangular shape to correspond to the shape of the condensing pump assembly **70**. The heater box **60** is sized to receive the pump assembly **70** such that it is substantially enclosed within the heater box **60**. More specifically, the condensate pump assembly **70** preferably slides freely into and out of the heater box **60** within the housing **22**. The pump assembly **70** is designed to sit within the housing without contacting the heater **50** on the bottom wall **66** of the heater box **60**. The heater **50**

is held by the heater cover 52 below and slightly spaced from the condensate pump assembly 70. (36) Referring to FIG. 19, another embodiment for use with the condensate pump assembly 70 comprises an uninsulated inner metal box 72 configured to receive the condensate pump 70. The inner box 72 defines a galvanized metal compartment serving multiple purposes, including accommodating all electrical components and connections, as well as supporting the silicone patch heater 50 and porous heater cover 52. The inner box 72 has outwardly extending flanges 74 along its upper edges. The box 72 is sized to slide into and out of the housing 22. When the box is received within the housing 22, the flanges 74 overhang the edges of the walls of the housing 22 such that the box 72 and pump assembly 70 are suspended above and spaced from the heater 50. In this arrangement, the box 72 can be configured to fit into or around most furnace locations as needed. A conventional fastener, such as a screw, may be used to secure the box in the selected position. When the box 72 is in position, the condensate pump assembly 70 or other furnace component or fluid location, such as a trap, is protected against freezing. A PVC drain line 44 (not shown) may exit the box 72 or the housing 22 for carrying fluid away. A similar “slide-in” arrangement would be suitable in other furnace locations.

(37) Although the present invention has been shown and described in considerable detail with respect to only a few exemplary embodiments thereof, it should be understood by those skilled in the art that it is not intended to limit the invention to the embodiments since various modifications, omissions and additions may be made to the disclosed embodiments without materially departing from the novel teachings and advantages of the invention, particularly in light of the foregoing teachings.

Claims

1. An apparatus for protecting from freezing fluid flowing from a conduit outlet above ground level on an exterior wall of a building, the freeze protection apparatus comprising: (a) a condensing gas furnace providing a continuous flow of condensate fluid through the conduit outlet; (b) a heater box having a longitudinal dimension, the heater box including a top wall, a front wall, and a plurality of side walls interconnecting the top wall and the front wall, wherein the heater box partially defines an enclosed area sized for covering the conduit outlet and having a rear aperture and a bottom aperture; (c) a housing adapted for receiving the heater box, the housing including a top wall, having an interior surface and an inner edge, a front wall having an interior surface, a bottom edge, and a lateral dimension, and, a pair of opposed side walls, each side wall having an interior surface, an inner edge, a bottom edge and a depth dimension, the side walls interconnecting the top wall and the front wall, wherein the housing has a longitudinal dimension greater than the longitudinal dimension of the heater box and partially defines an enclosed area having openings at a rear aperture for passing the heater box; (d) thermal insulating material lining the interior surface of the walls of the housing; (e) a thermostatically controlled electric heater mounted to the heater box for maintaining the enclosed area of the heater box above freezing temperature to prevent the condensate fluid from freezing; and (f) means for mounting the housing to the exterior wall adjacent the conduit outlet, wherein heat is retained within the heater box to prevent fluid within the conduit outlet from freezing.
2. The freeze protection apparatus as recited in claim 1, further comprising a bottom wall spanning the side walls and the front wall inwardly longitudinally spaced from distal edges of the side walls and the front wall.
3. The freeze protection apparatus as recited in claim 2, wherein the bottom wall is perforated to allow fluid to pass through.
4. The freeze protection apparatus as recited in claim 1, wherein the electric heater comprises a silicone pad heater.
5. The freeze protection apparatus as recited in claim 1, wherein the inner edges of the side walls

and the top wall of the housing contact the exterior wall.

6. The freeze protection apparatus as recited in claim 5, wherein the bottom edges of the side walls and the front wall are configured to contact the ground at ground level below the conduit outlet for enclosing the conduit outlet within the enclosed area defined by the heater box, the exterior wall and the ground level.

7. The freeze protection apparatus as recited in claim 1, wherein the mounting means comprises flanges at the inner edges of the side wall and the top wall of the housing.

8. The freeze protection apparatus as recited in claim 1, wherein the mounting means comprises a hinge for hingedly securing the housing to the exterior wall for pivoting movement between an open position away from the conduit outlet and a closed position enclosing the conduit outlet.

9. The freeze protection apparatus as recited in claim 8, wherein the pivot axis of the housing is about a substantially horizontal axis.

10. An apparatus for protecting from freezing fluid flowing from a condensing gas furnace providing a continuous flow of condensate fluid through a trap, the freeze protection apparatus comprising: (a) a heater box having a longitudinal dimension, the heater box including a top wall, a front wall, and a plurality of side walls interconnecting the top wall and the front wall, wherein the heater box partially defines an enclosed area and having a rear aperture and a bottom aperture for receiving the trap into the enclosed area; (b) a housing adapted for receiving the heater box, the housing including a top wall having an interior surface and an inner edge, a front wall having an interior surface, a bottom edge and a lateral dimension, and a plurality of side walls, each side wall having an interior surface, an inner edge, a bottom edge and a depth dimension, the side walls interconnecting the top wall and the front wall, wherein the housing partially defines an enclosed area having openings at a rear aperture and a bottom aperture, and wherein the area of the bottom aperture defined by the lateral dimension of the front wall and the depth dimension of the side walls is configured for receiving the heater box; (c) thermal insulating material lining the interior surface of the walls of the housing; (d) a thermostatically controlled electric heater mounted to the heater box for maintaining the temperature of the enclosed area of the heater box above freezing temperature to prevent the condensate fluid from freezing; and (e) means for mounting the housing to the furnace adjacent the trap, wherein the inner edges of the side walls and the top wall of the housing contact the furnace for enclosing the trap within an enclosed space defined by the heater box, and wherein heat is retained within the heater box to prevent the condensate fluid from freezing.

11. The freeze protection apparatus as recited in claim 10, further comprising a bottom wall spanning the side walls and the front wall inwardly longitudinally spaced from their distal edges.

12. The freeze protection apparatus as recited in claim 11, wherein the bottom wall is perforated to allow fluid to pass through.

13. The freeze protection apparatus as recited in claim 10, wherein the heater comprises a silicone pad heater.

14. The freeze protection apparatus as recited in claim 10, further comprising flanges at the inner edges of the side wall and the top wall.

15. The freeze protection apparatus as recited in claim 10, further comprising a hinge for hingedly securing the housing to an exterior wall of the furnace for pivoting movement between an open position away from the trap and a closed position enclosing the trap.

16. The freeze protection apparatus as recited in claim 15, wherein the pivot axis of the housing is about a substantially horizontal axis.
